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| miR-92 | *-AUUGCA-* | *miR-92* |

*Experiment 1*

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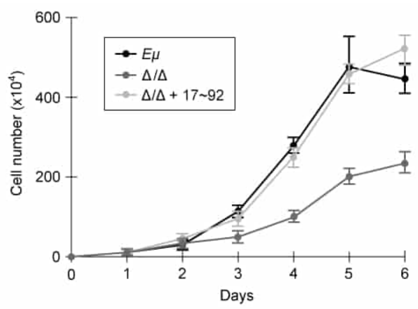


**Figure 1** Growth curves of *Eμ* cells, Δ/Δ cells, and Δ/Δ + *17~92* cells (error bars = standard deviation)

*Experiment 2*

To assess the relative contribution of each miRNA to the oncogenic ability of the *miR-17~92* cluster, Δ/Δ cell lines were transduced with an empty vector or with one of several *miR-17~92* mutant allele constructs that were engineered to lack expression of miRNA from at least one of the four families in the table. In each of the constructs, the family that was knocked out of the vector is indicated by the Δ symbol. After verifying that the transduced cells correctly expressed the desired miRNA, the fraction of apoptotic cells in each line was measured (Figure 2).

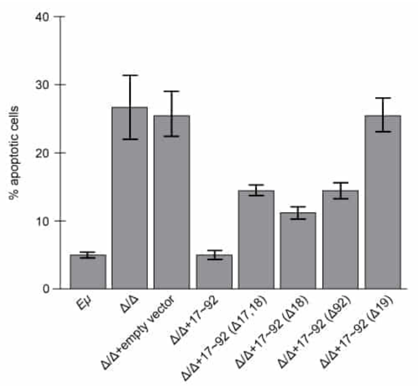


**Figure 2** Percentage of apoptotic cells in *Eμ* cells and Δ/Δ cells transduced with the indicated *miR-17~92* construct (error bars = standard deviation)

When expressed, the most likely function of the products generated by the *miR-17~92* gene cluster is to:

- ○ A. assist with the primary enzymatic functions of the ribosome. [7%]
- ○ B. facilitate the processing of pre-mRNA in the cell nucleus. [20%]
- ✔ ◉ C. interfere with gene expression by binding transcripts with complementary nucleotide sequences. [66%]
- ○ D. prevent small nuclear ribonucleoproteins from forming a large RNA-protein complex. [5%]

Correct
67% answered correctly

**Explanation:**

| Coding RNA | |
|---|---|
| **Messenger RNA (mRNA)** | • Is translated into protein by ribosomes |
| **Noncoding RNA** | |
| **Ribosomal RNA (rRNA)** | • Associates with specific proteins to form ribosomes |
| **Transfer RNA (tRNA)** | • Pairs mRNA codons with specific amino acids during translation |
| **Small nuclear RNA (snRNA)** | • Associates with specific proteins to form small nuclear ribonucleoproteins (snRNPs), the building block of spliceosomes |
| **Small interfering RNA (siRNA)** | • Functions in RNA interference<br>• Binds complementary mRNA and signals for its degradation |
| **MicroRNA (miRNA)** | • Functions in RNA interference<br>• Binds target complementary sequence on mRNA molecule to silence gene expression |

MicroRNAs **(miRNAs)** are small, noncoding eukaryotic or viral RNA molecules that bind complementary sequences on target messenger RNA (mRNA) molecules, consequently inhibiting their expression. miRNAs, such as those encoded by the *miR-17~92* gene cluster described in the passage, act to **silence gene expression** at the translational level. Following binding, miRNA-mediated silencing occurs either by promoting endonuclease activation and subsequent cleavage of the target mRNA or by preventing the target mRNA from binding to ribosomes (blocking translation). As a result, the ***miR-17~92*** **gene** cluster products are miRNAs that **interfere** with the **expression of specific genes** by binding target transcripts containing complementary nucleotide sequences.

**(Choice A)** Ribosomal RNA (rRNA) pairs with specific proteins to form the ribosome, a molecular complex that brings mRNA and tRNA together to enzymatically manufacture polypeptides during translation.

**(Choices B and D)** Small nuclear RNAs (snRNAs), a group of molecules confined to the nucleus, function in the splicing of pre-mRNA. snRNAs associate with a specific set of proteins to form small nuclear ribonucleoproteins (snRNPs). Various snRNPs then assemble with other proteins to form a spliceosome, the large protein–RNA complex involved in splicing.

**Educational objective:**
MicroRNAs (miRNAs), an example of noncoding RNA, silence gene expression at the translational level. miRNAs bind complementary sequences on target messenger RNA (mRNA) molecules, consequently inhibiting expression of the target mRNA by either blocking its translation or marking it for degradation.

**Time spent:**0 sec
**Subject:**
Biology

**QID:**400428
**System:**
Molecular Biology

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*Experiment 1*

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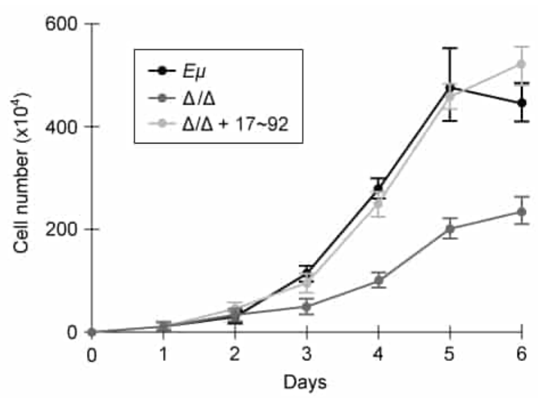



**Figure 1** Growth curves of *Eμ* cells, Δ/Δ cells, and Δ/Δ + *17~92* cells (error bars = standard deviation)

*Experiment 2*

To assess the relative contribution of each miRNA to the oncogenic ability of the *miR-17~92* cluster, Δ/Δ cell lines were transduced with an empty vector or with one of several *miR-17~92* mutant allele constructs that were engineered to lack expression of miRNA from at least one of the four families in the table. In each of the constructs, the family that was knocked out of the vector is indicated by the Δ symbol. After verifying that the transduced cells correctly expressed the desired miRNA, the fraction of apoptotic cells in each line was measured (Figure 2).

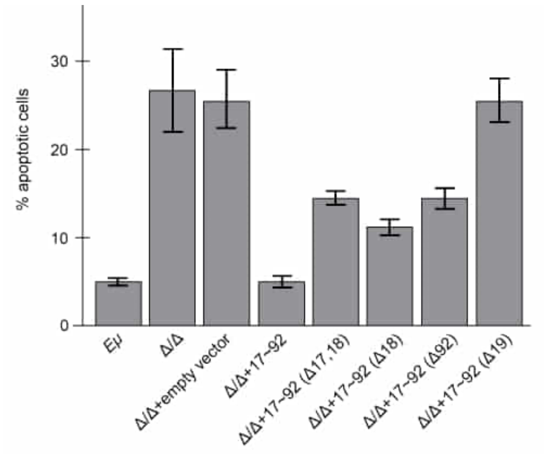
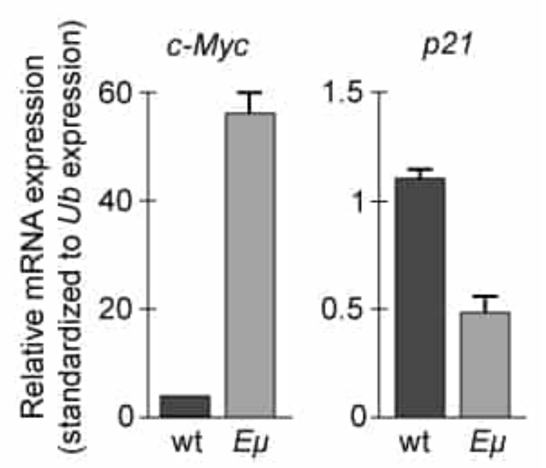



**Figure 2** Percentage of apoptotic cells in *Eμ* cells and Δ/Δ cells transduced with the indicated *miR-17~92* construct (error bars = standard deviation)

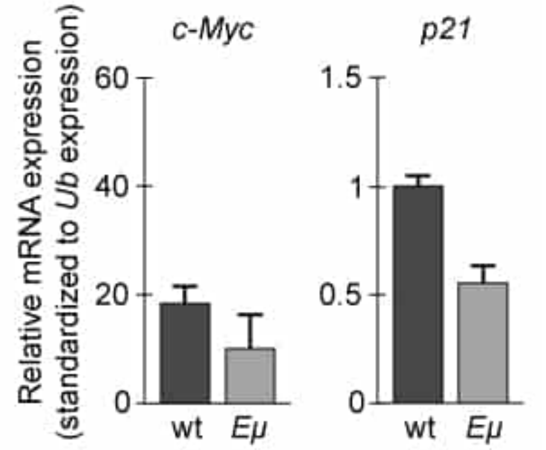
If changes in *p21* expression contribute to the development of B-cell lymphoma in *Eμ-Myc* mice, what is the expected result of real-time PCR analysis of *c-Myc* and *p21* expression in wild-type (wt) and *Eμ-Myc* (Eμ) mice? (Note: mRNA levels are standardized to the expression of *ubiquitin* (*Ub*) mRNA, which is not regulated by *c-Myc*.)

✔ ◉ A.



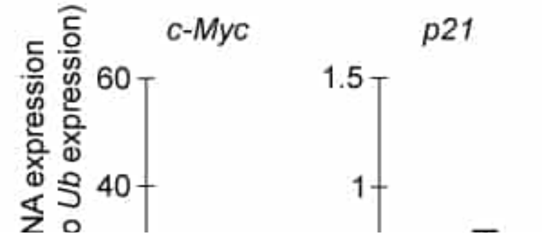

[55%]

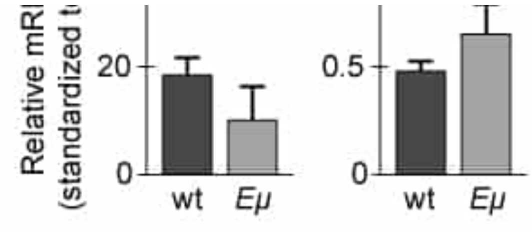
○ B.




[11%]

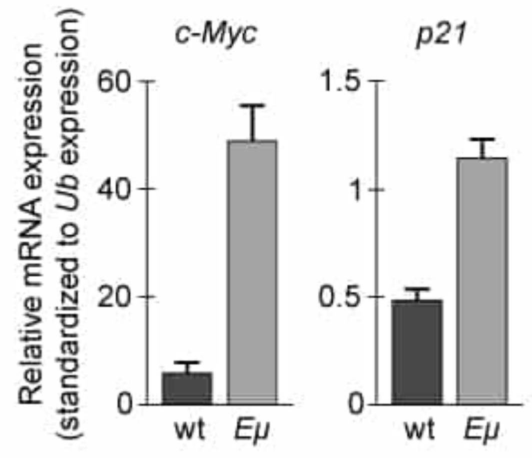
○ C.






[6%]

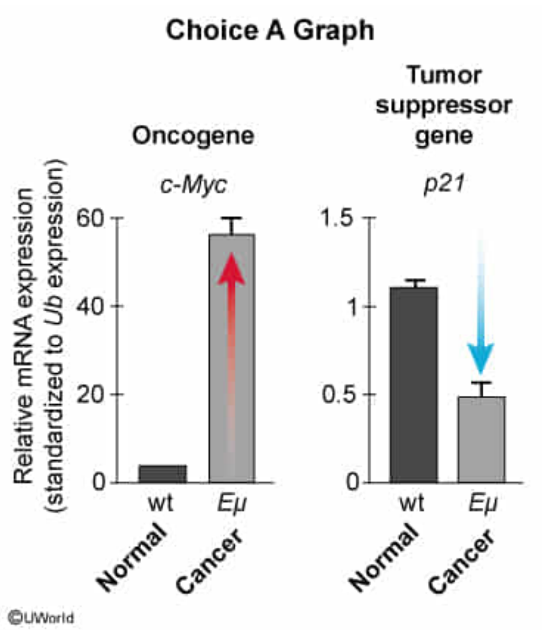
◯ D.




[26%]

Correct
56% answered correctly

**Explanation:**




Standard polymerase chain reaction (PCR) measures DNA amplification after all the thermal cycles are complete, whereas real-time PCR quantifies product amplification as the reaction progresses in real time. The researchers assessed expression of *c-Myc* and *p21* in wild-type (normal) and *Eμ-Myc* (Eμ) mice using real-time PCR.

According to the passage, ***c-Myc* overexpression** leads to **early cancer development** (ie, B-cell lymphoma) in *Eμ* mice. Therefore, *c-Myc* is most likely an oncogene, a mutated or overexpressed gene that induces uncontrolled cell growth by promoting cell cycle progression or inhibition of apoptosis. As a result, **higher *c-Myc* levels in Eμ mice** than in wild-type mice would likely promote the cancerous cell growth observed in the Eμ mice.

Based on the passage, regulation of *p21* expression by *c-Myc* may contribute to B-cell lymphoma development in *Eμ* mice. The passage states that *p21* **inhibits cell cycle progression**, a key feature of a **tumor suppressor gene**. Tumor suppressor genes regulate DNA repair by repressing or pausing the cell cycle to ensure that only normal cells proceed to the division stage and induce programmed cell death if repair fails. As a result of higher *c-Myc* expression (and consistent with greater B-lymphoma development), ***p21* levels** would likely be **lower in *Eμ* mice** compared to wild-type mice.

**(Choice B)** *p21* expression levels are correct in this graph; however, *c-Myc* expression should be increased (*not decreased*) in *Eμ* mice compared to wild-type mice.

**(Choice C)** Compared to wild-type mice, *c-Myc* should be expressed *more* abundantly and *p21* should be expressed *less* abundantly in *Eμ* mice.

**(Choice D)** Although *c-Myc* expression levels are correct in this graph, *p21* levels should be lower (*not higher*) in *Eμ* mice compared to wild-type mice.

**Educational objective:**
In the setting of cancer, oncogenes are mutated or expressed at abnormally high levels and contribute to cancer development by promoting cell growth/proliferation or suppressing apoptosis. In contrast, tumor suppressor genes, which induce programmed cell death, are inhibited in cancerous cells.

| | |
|---|---|
| **Time spent:**0 sec | **QID**:400429 |
| **Subject:** Biology | **System:** Molecular Biology |

Overexpression of the *miR-17~92* gene cluster occurs in many human cancers involving *c-Myc*, a gene coding for a transcription factor that regulates expression of proteins such as p21, an inhibitor of cell cycle progression. *miR-17~92* encodes six distinct microRNAs (miRNAs) that can be divided into four families based on sequence identity at positions 2-7 (Table 1).

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*Experiment 1*

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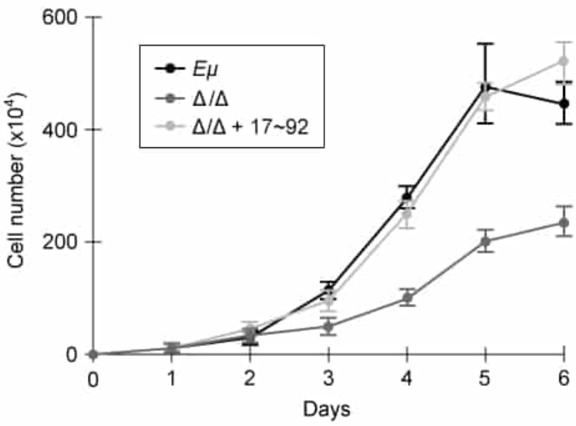



**Figure 1** Growth curves of *Eμ* cells, Δ/Δ cells, and Δ/Δ + *17~92* cells (error bars = standard deviation)

*Experiment 2*

To assess the relative contribution of each miRNA to the oncogenic ability of the *miR-17~92* cluster, Δ/Δ cell lines were transduced with an empty vector or with one of several *miR-17~92* mutant allele constructs that were engineered to lack expression of miRNA from at least one of the four families in the table. In each of the constructs, the family that was knocked out of the vector is indicated by the Δ symbol. After verifying that the transduced cells correctly expressed the desired miRNA, the fraction of apoptotic cells in each line was measured (Figure 2).

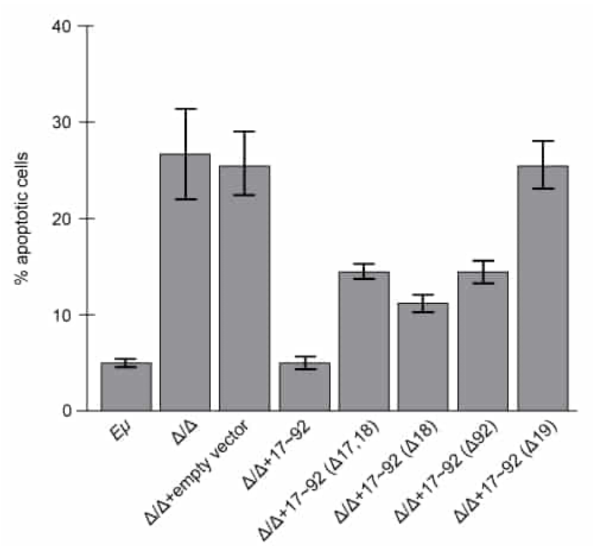



**Figure 2** Percentage of apoptotic cells in *Eμ* cells and Δ/Δ cells transduced with the indicated *miR-17~92* construct (error bars = standard deviation)

Which of the following conclusions about *c-Myc* and the *miR-17~92* gene cluster is (are) most consistent with the information in the passage?

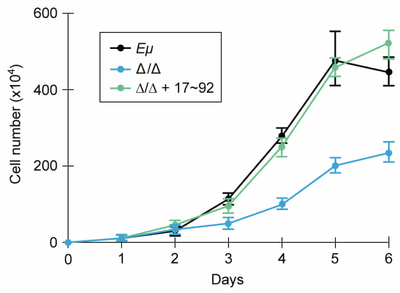
I. *c-Myc* induces *miR-17~92* expression.
II. *c-Myc* knockout promotes *miR-17~92* function.
III. *miR-17~92* suppresses apoptosis.

- A. I only [12%]
- B. I and II only [11%]
- C. I and III only [68%]
- D. I, II, and III [7%]

Correct
69% answered correctly

**Explanation:**

Figure 1 shows the growth curves (cell number) of three different cell lines: *Eμ-Myc* cells (*Eμ*), *Eμ-Myc* cells with miR-17~92 knocked out (Δ/Δ), and *Eμ-Myc* cells with *miR-17~92* knocked out but in which *miR-17~92* was reintroduced (Δ/Δ + *miR-17~92*). The results of Figure 1 can be interpreted as shown below:




| Cell line | *c-Myc* expression | *miR-17~92 cluster* expression | Cell number |
|---|---|---|---|
| *Eμ* | Overexpressed | Normal | +++ |
| Δ/Δ | Overexpressed | Absent (knocked out) | + (decreased) |
| Δ/Δ + 17~92 | Overexpressed | Normal (restored) | +++ (restored) |



Figure 1 shows that cell proliferation is increased when *c-Myc* and the *miR-17~92* cluster are expressed (*Eμ-Myc*). However, *c-Myc* expression alone (ie, *Eμ-Myc* with the *miR-17~92* cluster knocked out [Δ/Δ]) results in decreased cell number, indicating that expression of *c-Myc only* does *not* likely promote increased cell proliferation (tumor development). This is consistent with the conclusion that ***c-Myc*** could **promote tumor development by first inducing the *miR-17~92* cluster**, which in turn facilitates cell proliferation. This is evidenced by the restoration of increased cell number on reintroduction of the *miR-17~92* cluster **(Number I)**.

In addition, according to Figure 2, *Eμ-Myc* cells express *miR-17~92* normally and show only ~5% apoptotic cells. However, when *miR-17~92* is knocked out in *Eμ-Myc* cells (Δ/Δ), the percentage of apoptotic cells is *increased* to ~28%. Therefore, it can be concluded that ***miR-17~92* expression suppresses apoptosis (Number III)**.

**(Number II)** In the passage, overexpression (*not* knockout) of *c-Myc* in *Eμ-Myc* cells was studied. Furthermore, Figure 1 shows that higher levels of cell proliferation occur *only* when *both c-Myc* and the *miR-17~92* cluster are present. Therefore, *knocking out c-Myc* would likely inhibit (*not* promote) *miR-17~92* function, ultimately *decreasing* cell proliferation and cancer development.

**Educational objective:**

To analyze whether one gene regulates another, researchers can assess variables of interest (eg, cell growth, apoptosis) using engineered cell lines in which one of the target genes is expressed and the other gene of interest is silenced and then reintroduced.

**Time spent:**0 sec
**Subject:**
Biology

**QID**:400430
**System:**
Molecular Biology

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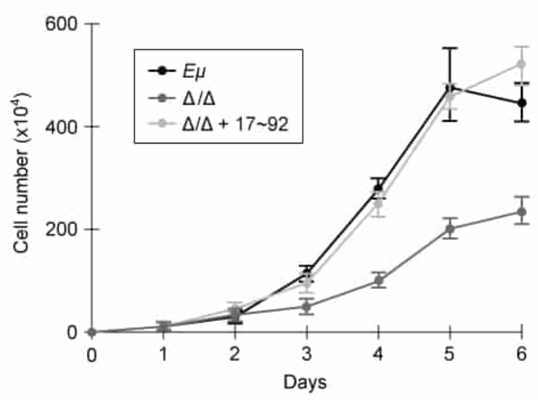


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*Experiment 2*

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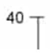


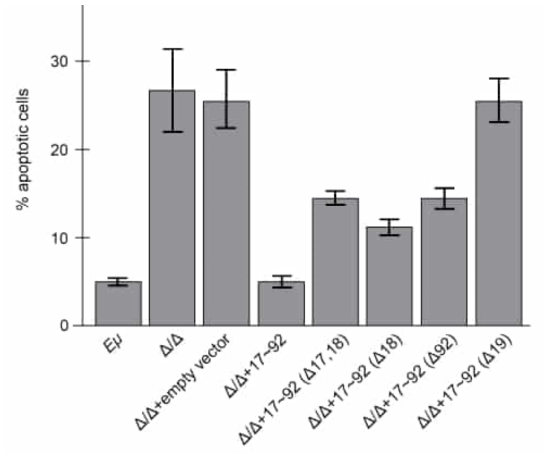


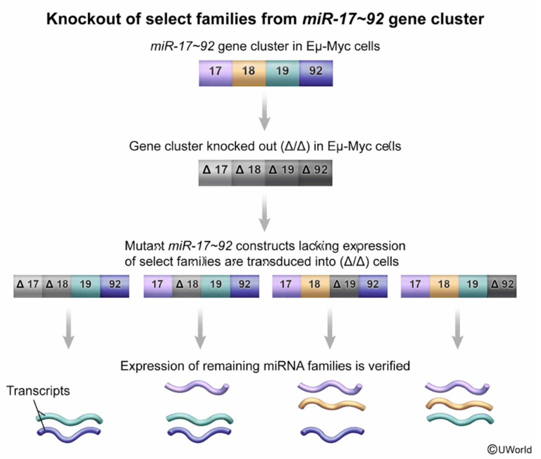

**Figure 2** Percentage of apoptotic cells in *Eμ* cells and Δ/Δ cells transduced with the indicated *miR-17~92* construct (error bars = standard deviation)

As described in the passage, verification that the transduced cells expressed the desired miRNA was an essential experimental step to ensure that:

- A. lack of expression of one or more miRNA encoded by the *miR-17~92* cluster did not affect transcription of pro-apoptotic genes. [26%]
- B. transduction of tested cells with mutant miR-17~92 alleles did not induce the development of B-cell lymphoma. [9%]
- ✔ C. deletion of one or more miRNA encoded by the *miR-17~92* cluster did not affect expression of the remaining miRNAs encoded by the constructs. [48%]
- D. expression of one or more mutant miR-17~92 alleles did not provide the transduced cells with a selective growth advantage. [15%]

Correct
49% answered correctly

**Explanation:**




During an experiment, controls are used to demonstrate that changes observed in the dependent variable were due *only* to changes in the manipulated independent variable (ie, that results were *not* impacted by confounding variables).

In *Experiment 2*, verifying that each mutant construct works properly and expresses only the desired families serves as an experimental control. In Figure 2, the independent variable (treatment) is the type of vector transduced into *Eμ-Myc* cells with *miR-17~92* knocked out (*Δ/Δ*), and the dependent variable is the percentage of apoptotic cells. The purpose of measuring apoptosis under these conditions was to determine precisely which miRNA families contribute the most to *miR-17~92* function.

To reliably draw such conclusions, it was necessary to first ensure that **deletion of one or more miRNAs encoded by** the ***miR-17~92*** cluster **did not affect expression**/processing of the **remaining miRNAs** encoded by the mutant constructs (ie, that apoptosis was affected *only* by deletion of the *selected* miRNAs).

**(Choice A)** Although researchers assessed the fraction (percentage) of apoptotic cells in samples with different genotypes, the transcription of pro-apoptotic genes was *not* directly measured.

**(Choice B)** According to the passage, all the cell lines tested were derived from *Eμ-Myc* mice, which *already* exhibited early development of diffuse B-cell lymphoma.

**(Choice D)** The purpose of the experiment was to determine the extent to which introduction of constructs containing one or more *miR-17~92* alleles *did* affect cell growth in the lines tested.

**Educational objective:**
Experimental controls are used to demonstrate that the dependent variable in an experiment was affected only by the independent variable and not by confounding factors.

| | |
|---|---|
| **Time spent:**0 sec | **QID**:400431 |
| **Subject:** Biology | **System:** Molecular Biology |